INTERNSHIP REPORT

ITETE BUNA & RESTAURANT PLATFORM

Digital QR Menu & Operations Platform • Physical to Digital Transition & Staff RBAC

PROJECT & INTERNSHIP METADATA

Candidate / Intern:	Software Engineering Intern
Program of Study:	B.Sc. in Computer Science / Software Engineering
Brand Identity:	ITETE BUNA (Authentic Single-Origin & Dining)
Live Customer Menu:	https://restaurant-menu-qr-code-customer.vercel.app
Core Mission:	Physical to Digital QR Menu Migration & Staff RBAC
Host Project:	Restaurant Menu QR Code Monorepo
Tech Stack:	React 18, Vite 5, Tailwind CSS, Framer Motion, Axios
Key Roles Supported:	Administrator (Full CRUD/Settings) vs. Waiter (Live Floor Ops)

Confidential & Academic Final Project Submission

Generated: 8/30/2026

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1. Executive Summary & Core Mission

The primary purpose of this project is the digital transformation of traditional restaurant operations: replacing physical paper menus with an interactive, mobile-first QR table ordering suite. When customers scan the table QR code, all restaurant offerings (food, hot/cold drinks, appetizers, desserts, daily specials) appear instantly with real-time stock availability. The platform equips administrators with full CRUD and hide/show controls, while granting waitstaff dedicated operational access.

2. Introduction: Transitioning Physical Menus to Digital

2.1 Limitations of Physical Paper Menus

Physical menus suffer from high reprinting costs, inability to reflect out-of-stock items, slow table turnarounds, and lack of dietary/allergen transparency. Whenever dishes or prices change, physical menus must be redesigned, printed, and laminated.

2.2 Digital QR Solution & Academic Objectives

This project solves these challenges by providing instant browser-based QR menus with live availability toggling, dietary filters, and direct kitchen order dispatching. The project reinforces computer science coursework in distributed web architecture, role-based security, state synchronization, and human-computer interaction.

3. Company Background & Hospitality Tech Profile

The host organization develops Hospitality Technology (FoodTech) and SaaS products, delivering contactless dining menus, real-time Kitchen Display Systems (KDS), table management, and analytics to hospitality venues.

4. Customer QR Menu Discovery & Ordering Experience

4.1 QR Scanning & Dynamic Table Routing

Guests scan a physical table stand QR code with their smartphone camera. The scanner immediately routes to the table menu (/menu/qr/:shortId) linking the table name (e.g. Table 3 Patio) without requiring user login or app installation.

4.2 Comprehensive Food & Beverage Catalog

- **Full Menu Visibility:** Displays all food, drinks, desserts, and specials organized by clear category tabs.
- **Dietary & Allergen Filtering:** One-touch filter pills for Vegetarian (Veg), Vegan, Gluten-Free (GF), Halal, and Spicy levels.
- **Interactive Cart Drawer:** Guests customize quantities, attach special kitchen notes (e.g. no onions), and submit orders.
- **Table Service Actions:** Floating action pill with instant "Call Waiter" and "Request Bill" (Card / Cash) alerts.
- **Guest Wi-Fi & Language Switcher:**
  One-tap guest Wi-Fi password copy and English / Amharic / French / Spanish translation.

5. Admin & Waiter Operational Control (RBAC)

5.1 Full Admin Control: Add, Edit, Delete, and Hide/Show (Stock Toggle)

- **Add & Edit Dishes:**
  Admins can upload multiple dish images, configure prices, discount rates, spicy levels, and cooking times.
- **Instant Hide/Show (Stock Control):**
  Single-click switch on MenuCard toggles items between Active (In-Stock) and Hidden (Out-of-Stock). Hidden items disappear from customer menus immediately.
- **Permanent Deletion:** Safely remove discontinued items and categories with confirmation safeguards.

5.2 Role-Based Access: Admin vs. Waiter / Staff Roles

Role-Based Capabilities Matrix

- Admin / Manager Role: Full control over Menu CRUD, Branding Settings, Currencies, Tax/Service fees, and Analytics.
- Waiter / Staff Role: Dedicated access to Live Orders queue, Waiter Call notifications, Bill Requests, and Stock Toggles.

5.3 Live Kitchen Display & Printable Table Stands

Orders dispatched by guests appear in real time on the Live Kitchen Queue (/dashboard/orders) with Kanban status progression (Pending -> In Kitchen -> Ready -> Served). The QR Manager (/dashboard/qr) generates print-ready A5 PDF acrylic table stands with high-res QR codes and Wi-Fi credentials.

6. Technical Skills & Engineering Knowledge Utilized

- **Frontend Architecture:** React 18, Vite 5, Tailwind CSS 3, Framer Motion 10, JavaScript ES6+.
- **Data Transport & State:** Axios with JWT interceptors, React Context API, and cross-tab storage broadcasting.
- **Document Generation:** jsPDF and html2canvas for vector A5 acrylic table stand generation.
- **Agile & Version Control:** Git, GitHub feature branches, pull requests, semantic commits, and sprint retrospectives.

7. Collaboration, Agile Processes & Teamwork

Collaborated in a 2-week Agile Scrum cadence with daily standups, code reviews, and cross-functional design handoffs. Ensured zero build regressions and maintained 100% clean production compilations across all workspaces.

8. Project Highlights & Key Case Studies

Case Study 1: Real-Time Table Ordering & Live Kitchen Queue

Problem: Paper menus cause order errors, miscommunicated dietary requests, and billing bottlenecks. Solution: Built an end-to-end table ordering pipeline with an animated cart drawer, live status progression, and instant table assistance banner alerts.

Case Study 2: White-Label Customizer & Acrylic Table Stand PDF Generator

Problem: Restaurants need custom branding and physical acrylic QR table tents. Solution: Built a Settings panel (/dashboard/settings) supporting 7 theme palettes, multi-currency formatting (ETB, $, EUR, GBP), and an automated single-click A5 PDF table tent export.

9. Challenges Faced, Root Cause Analysis & Solutions

- **Vite 5 / Node Toolchain Alignment:**
  Resolved alpha dependency native binding issues by standardizing all workspaces to stable Vite 5 and npm workspace scripts.
- **Cross-Tab Reactive Sync:**
  Employed window storage listeners and custom dispatch events to broadcast stock updates and new orders across open tabs in real time.
- **High-DPI PDF QR Printing:**
  Configured html2canvas with 3x scale multiplier and high error correction in qrcode.react for sharp table tent printouts.

10. Organizational Value Added, Handover & Conclusion

The digital QR menu system successfully eliminates paper menu reprinting expenses, increases table turnover velocity, and equips both managers and waiters with a robust operational toolset.

Summary of Project Deliverables

- Full Monorepo Codebase: Clean React 18 / Vite source code in apps/admin, apps/customer, and packages/shared.
- Digital Menu Catalog: High-speed mobile-first menu displaying all food, drink, and dessert categories.
- Admin Control Suite: Comprehensive Add, Edit, Delete, and In-Stock Hide/Show toggle tools.
- Waiter Operational View: Live Kitchen Display Kanban, table service alerts, and status pipelines.
- Complete Documentation: Formatted README.md, INTERNSHIP_REPORT.md, and Project_Documentation.pdf.